



UNIVERSITY of
RWANDA

DEPARTMENT OF PHYSICS

FIRST TRIMESTER EXAMINATIONS – ACADEMIC YEAR 2019-2020

DEPARTMENT: PHYSICS

LEVEL OF STUDY: Year 1

STUDY GROUP: CE_CE, CE_EE, CE_GE, CE_SE, CE_WE, SGE , 1

**CS_Huye , MEE_PROD_Gako, IS_Huye , IT_Huye , CS_Musanze, EEE_EPE,
EEE_ETE**

MODULE CODE: PHY1163

MODULE NAME: Physics for Engineers I

DATE: 17/02/2020

DURATION: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This paper contains 5 questions to be answered to in 2 Sections: Section A and Section B.
2. Section A is compulsory; you must answer to all the questions.
3. Section B allows a choice; select any 2 questions from the 3 questions given and answer to them.
4. Write both your Registration and your Examination Numbers on the Examination Booklets provided.
5. Write all your answers only in the examination booklet provided.
6. Do not write on this question paper.
7. Start each question on a fresh page.
8. No unauthorized material is allowed in the examination room.

SECTION A: COMPULSORY QUESTIONS / 20 marks

Question N°1/10 marks

- a) i) Using the method of dimensional analysis, show that the period of oscillation of a mass suspended on spiral spring is proportional to the spring constant and the mass according to the following equation **1.5 marks**

$$T^2 \sim \frac{1}{n}$$

$$= \text{kg} \cdot \text{rad/s}^2$$

$$T^2 \sim$$

$$T \sim \sqrt{\frac{K}{m}}$$

ii) State the Newton's laws of motion **1.5 marks**

iii) What are conditions of static equilibrium for a rigid body **1 marks**

iv) Differentiate between a laminar flow and a turbulent flow **1 marks**

b) i) A block of mass $m = 1.5 \text{ kg}$ hangs at the end of a (weightless) cord suspended from the ceiling. What is the tension in the cord, and with what force does the cord pull down on the ceiling. **2 marks**

ii) A metal rod that is 4.00 m long and 0.50 cm^2 in cross - sectional area is found to stretch 0.20 cm under a tension of 5000 N . What is Young's modulus for this metal? **1.5 marks**

c) When a body of unknown mass is attached to an ideal spring with force constant 120 N/m , it is found to vibrate with a frequency of 6.00 Hz . Find (i) the period of the motion; (ii) the angular frequency; (iii) the mass of the body. **1.5 marks**

Question N^o2/10 marks

a) An automobile initially moving at 30 m/s accelerates uniformly at 5 m/s^2 . How fast is it moving after 3 s ? **1 marks**

b) A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0 \text{ m/s}$ at an angle $\alpha = 53.1^\circ$. **3 marks**

(i) Find the position of the ball and its velocity (magnitude and direction) at $t = 2.00 \text{ s}$.

(ii) Find the time when the ball reaches the highest point of its flight, and its height at this time.

(iii) Find the horizontal range.

c) At a certain point in a horizontal pipeline the water's speed is 2.50 m/s and the gauge pressure is $1.80 \times 10^4 \text{ Pa}$. Find the gauge pressure at a second point in the line if the cross-sectional area at the second point is twice that at the first. **1 marks**

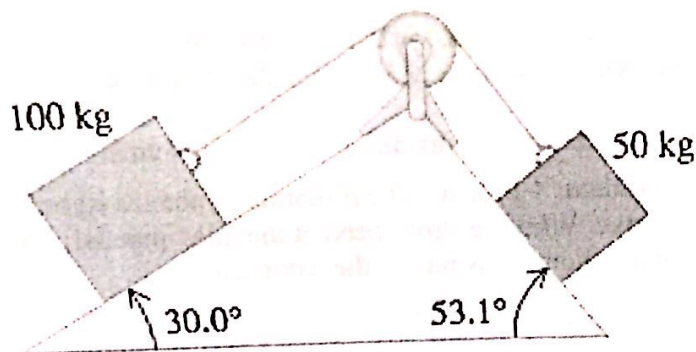
SECTION B: FROM THIS SECTION, CHOOSE ONLY TWO QUESTIONS
marks / 30

Question N^o3 / 15 marks

a) A block of mass $m = 6.0 \text{ kg}$ slides from rest on a horizontal frictionless surface under the action of a force of 70 N in a direction 50° above the positive x-axis. How much work does the force after the end of 8 s . **6 marks**

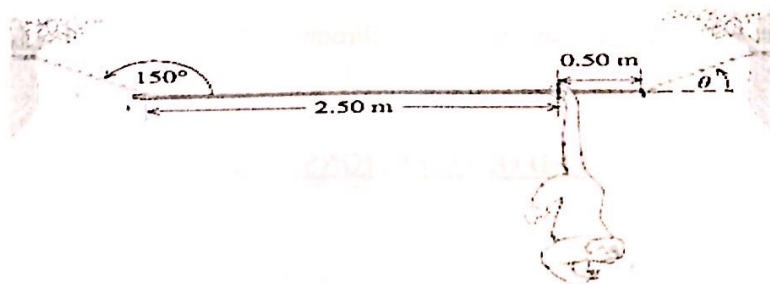
b) Two blocks connected by a cord passing over a small, frictionless pulley rest on frictionless planes **9 marks**

- Which way will the system move when the blocks are released from rest?
- What is the acceleration of the blocks?
- What is the tension in the cord?

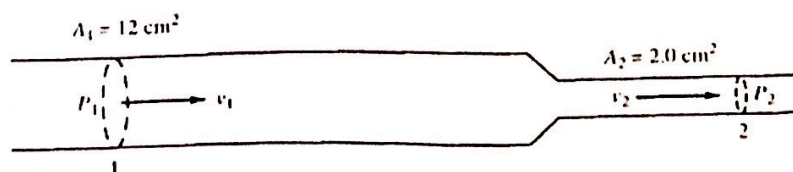


Question N^o4 / 15 marks

a) A 3.00 m long, 24 kg , uniform rod at the zoo is held in a horizontal position by two ropes at its ends. The left rope makes an angle of 150° with the rod and the right rope makes an angle θ with the horizontal. A 9 kg howler monkey hangs motionless 0.50 m from the right end of the rod as he carefully studies you. Calculate the tensions in the two ropes and the angle θ . First make a free body diagram of the rod. **7 marks**



b) Consider a section of a horizontal pipe with water flowing through it as shown in the figure below



- i) If the speed of the water at point 1 is $v_1 = 2.0 \text{ m/s}$, find the velocity v_2 at point 2. **4marks**
- ii) If the pressure of the fluid at point 1 is $P_1 = 2.0 \times 10^5 \text{ Pa}$, find the pressure P_2 at point 2. **4marks**

Question N°5 / 15 marks

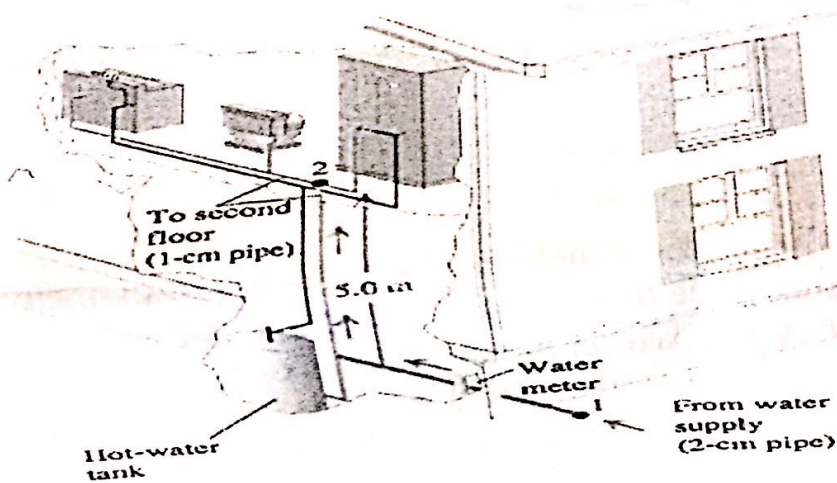
a) A tennis ball on Mars, where the acceleration due to gravity is $0.379g$ and air resistance is negligible, is hit directly upward and returns to the same level 8.5 s later.

(i) How high above its original point did the ball go? **3 marks**

(ii) How fast was it moving just after being hit? **3 marks**

(iii) Sketch graphs of the ball's vertical position, vertical velocity, and vertical acceleration as functions of time while it's in the Martian air. **3 marks**

b) Water enters a house through a pipe with an inside diameter of 2.0 cm at an absolute pressure of $4.0 \times 10^5 \text{ Pa}$ (about 4 atm). A 1.0 cm -diameter pipe leads to the second-floor bathroom 5.0 m above. When the flow speed at the inlet pipe is 1.5 m/s , find the flow speed, pressure, and volume flow rate in the bathroom.



What is the water pressure in the second-story bathroom of this house? **6marks**

END OF QUESTIONS

$$\frac{1}{3} \times 10^5$$

$$= \frac{1}{2} \times 10^5$$